

Simulation 2 Full Output

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=====
Setting 1: Exponential survival + Exponential censoring
Survival: Exponential(rate=0.10)
Censoring: Exponential(rate=0.05)
tau = 5
True F(tau)    = 0.393469
True RMST(tau)= 3.934693
=====
```

```
-----
Setting = 1 | n = 200 | nrep = 20
Mean censoring rate = 0.3262 (sd = 0.0274 )
-----
```

THEORETICAL VALIDATION: $J_i - \theta_{\text{true}}$ vs ϕ_{hat}

F(tau):

	correlation	mse	intercept	slope
mean	1	0.0012	0.0089	1.0089
sd	0	0.0014	0.0345	0.0006

RMST(tau):

	correlation	mse	intercept	slope
mean	1	0.0145	0.0025	1.0065
sd	0	0.0228	0.1230	0.0002

NUMERICAL CHECK: $J_i - \theta_{\text{hat}}$ vs ϕ_{hat}

F(tau):

	correlation	mse	intercept	slope
mean	1	0	0	1.0089
sd	0	0	0	0.0006

RMST(tau):

	correlation	mse	intercept	slope
mean	1	1e-04	0	1.0065
sd	0	0e+00	0	0.0002

```
-----
Setting = 1 | n = 500 | nrep = 20
Mean censoring rate = 0.3367 (sd = 0.0206 )
-----
```

THEORETICAL VALIDATION: $J_i - \theta_{\text{true}}$ vs ϕ_{hat}

F(tau):

	correlation	mse	intercept	slope
mean	1	6e-04	-0.0115	1.0034
sd	0	1e-03	0.0223	0.0001

RMST(tau):

	correlation	mse	intercept	slope
mean	1	0.0049	0.0159	1.0026
sd	0	0.0069	0.0695	0.0001

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0034
sd	0	0	0	0.0001

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0026
sd	0	0	0	0.0001

Setting = 1 | n = 5000 | nrep = 20
Mean censoring rate = 0.3317 (sd = 0.0064)

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	1e-04	-0.0001	1.0003
sd	0	1e-04	0.0096	0.0000

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0.0010	-0.002	1.0003
sd	0	0.0014	0.033	0.0000

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0003
sd	0	0	0	0.0000

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0003
sd	0	0	0	0.0000

=====
Setting 2: Weibull survival + Uniform censoring
Survival: Weibull(shape=2, scale=8)
Censoring: Uniform(0, 12)
 $\tau = 5$
True F(τ) = 0.323366
True RMST(τ) = 4.418663
=====

```
-----
Setting = 2 | n = 200 | nrep = 20
Mean censoring rate = 0.5698 (sd = 0.0364 )
-----
```

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0.0014	0.0002	1.0121
sd	0	0.0021	0.0370	0.0010

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0.0040	-0.0227	1.0080
sd	0	0.0052	0.0600	0.0004

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0121
sd	0	0	0	0.0010

RMST(τ):

	correlation	mse	intercept	slope
mean	1	1e-04	0	1.0080
sd	0	0e+00	0	0.0004

```
-----
Setting = 2 | n = 500 | nrep = 20
Mean censoring rate = 0.5663 (sd = 0.0224 )
-----
```

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	7e-04	0.0043	1.0048
sd	0	9e-04	0.0267	0.0004

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0.0033	-0.0097	1.0032
sd	0	0.0038	0.0580	0.0001

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0048
sd	0	0	0	0.0004

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0032
sd	0	0	0	0.0001

```
-----
Setting = 2 | n = 5000 | nrep = 20
Mean censoring rate = 0.5703 (sd = 0.0076 )
-----
```

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0e+00	0.0016	1.0005
sd	0	1e-04	0.0066	0.0000

RMST(τ):

	correlation	mse	intercept	slope
mean	1	2e-04	-0.0040	1.0003
sd	0	2e-04	0.0138	0.0000

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0005
sd	0	0	0	0.0000

RMST(τ):

	correlation	mse	intercept	slope
mean	1	0	0	1.0003
sd	0	0	0	0.0000

```
=====
Setting 3: Log-logistic survival + Exponential censoring
Survival: Log-logistic(shape=2, scale=4)
Censoring: Exponential(rate=0.12)
tau = 5
True F(tau) = 0.609756
True RMST(tau)= 3.584222
=====
```

```
-----
Setting = 3 | n = 200 | nrep = 20
Mean censoring rate = 0.4248 (sd = 0.0399 )
-----
```

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

F(τ):

	correlation	mse	intercept	slope
mean	1	0.0014	-0.0144	1.0181
sd	0	0.0015	0.0336	0.0018

```
RMST(tau):
      correlation    mse intercept    slope
mean          1 0.0075      0.0231 1.0097
sd            0 0.0090      0.0837 0.0006
```

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

```
F(tau):
      correlation    mse intercept    slope
mean          1 1e-04          0 1.0181
sd            0 0e+00          0 0.0018
```

```
RMST(tau):
      correlation    mse intercept    slope
mean          1 3e-04          0 1.0097
sd            0 0e+00          0 0.0006
```

```
-----
Setting = 3 | n = 500 | nrep = 20
Mean censoring rate = 0.4171 (sd = 0.0192 )
-----
```

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

```
F(tau):
      correlation    mse intercept    slope
mean          1 0.0007      0.0017 1.0074
sd            0 0.0012      0.0275 0.0009
```

```
RMST(tau):
      correlation    mse intercept    slope
mean          1 0.0086     -0.0047 1.0039
sd            0 0.0105      0.0950 0.0003
```

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

```
F(tau):
      correlation mse intercept    slope
mean          1  0          0 1.0074
sd            0  0          0 0.0009
```

```
RMST(tau):
      correlation mse intercept    slope
mean          1  0          0 1.0039
sd            0  0          0 0.0003
```

```
-----
Setting = 3 | n = 5000 | nrep = 20
Mean censoring rate = 0.418 (sd = 0.0069 )
-----
```

THEORETICAL VALIDATION: J_i - θ_{true} vs ϕ_{hat}

```
F(tau):
```

	correlation	mse	intercept	slope
mean	1	1e-04	0.0010	1.0007
sd	0	1e-04	0.0093	0.0000

RMST(tau):

	correlation	mse	intercept	slope
mean	1	9e-04	-0.0063	1.0004
sd	0	9e-04	0.0303	0.0000

NUMERICAL CHECK: J_i - θ_{hat} vs ϕ_{hat}

F(tau):

	correlation	mse	intercept	slope
mean	1	0	0	1.0007
sd	0	0	0	0.0000

RMST(tau):

	correlation	mse	intercept	slope
mean	1	0	0	1.0004
sd	0	0	0	0.0000